
CA 2025 HW1

— RISC-V Assembly Code —

Agenda

- Assignment Introduction
 - Description
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 - Sample Code
- Grading Policy
- Submission
- Jupiter Installation

Assignment Introduction

- In this homework, you are going to use [Jupiter RISC-V simulator](#) to develop a simple calculator.
- After finishing this homework, you will be familiar with the usage of Jupiter RISC-V simulator, register definition, and some basic operations in RV32I Base Integer Instruction Set.



Description

- You are going to develop a simple calculator, which supports seven operations.
- Addition(0), subtraction(1), multiplication(2), integer division(3), minimum(4), power(5), and factorial(6).

Sample I/O

- Input file contains 3 lines, operand A, operation op, operand B, respectively. ($0 \leq A, B \leq 1024$, $op \in \{0, 1, 2, 3, 4, 5, 6\}$)
- Your program should output the correct result ($A \text{ op } B$).

```
> ./jupiter hw1.s
10
0
10
20

Jupiter: exit(0)
```

Add

```
> ./jupiter hw1.s
10
3
0
division by zero

Jupiter: exit(0)
```

Division

```
> ./jupiter hw1.s
2
5
5
32

Jupiter: exit(0)
```

Power

Sample Code

- In the sample code, you don't need to do I/O operations by yourself. A, op, B will be stored at register s0, s1, s2 registers.
- You need to store the result to register s3.

```
6 .text
7 __start:
8     # Read first operand
9     li a0, 5
10    ecall
11    mv s0, a0
12    # Read operation
13    li a0, 5
14    ecall
15    mv s1, a0
16    # Read second operand
17    li a0, 5
18    ecall
19    mv s2, a0
```

```
26 output:
27     # Output the result
28     li a0, 1
29     mv a1, s3
30     ecall
31
32 exit:
33     # Exit program(necessary)
34     li a0, 10
35     ecall
```

[Jupiter Ecalls](#)

Sample Code

- If $op=3$ and $B=0$, just jump to `division_by_zero_except`.

```
37 division_by_zero_except:
38     li a0, 4
39     la a1, division_by_zero
40     ecall
41     jal zero, exit
```

Sample Code

- You may finish operation implementations.

```
21 #####  
22 #  TODO: Develop your calculator #  
23 #                                #  
24 #####
```


Grading Policy

- Total 100%
 - For operations +, -, x, / and min, each has 4 test cases, 3 points per test case.
 - For operations ^ and !, each has 5 test cases, 4 points per test case.
- We will judge your program by running the following command:

```
$ jupiter [student_id].s < input_file
```

- Don't worry about overflow and underflow.
- No need to handle 0^0 .
- 10 points off per day for late submission.
- You will get 0 point for plagiarism.

Submission

- Due date: 9/30 23:59 (Tuesday)
- Late Submission: Accepted until 10/07 (Tue) 23:59
- Please rename your program [student_id].s and upload it to NTU COOL.
 - For example, if your student id is **b12345678**, your program file name should be **b12345678.s**. (alphabet must be in lower case!)

Jupyter Installation

- Please go to jupyter [release page](#) to download the latest version (v3.1) according to your operating system.

Jupyter v3.1 Latest

Change Log

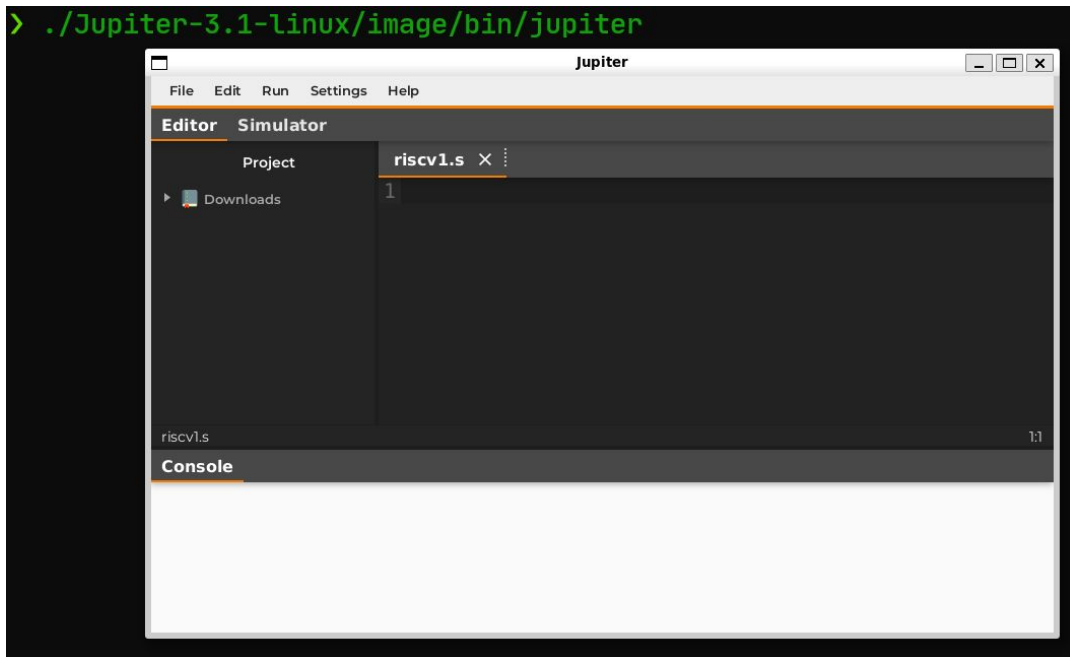
- Fix typo in Assembler.java, use data() instead of bss()
- Fix align method in LinkedProgram.java
- Remove explicit allocation with zeros in allocateBytesFromHeap for performance
- Add `-cache` option in CLI mode and Enable Cache Simulation setting in GUI mode
- Add deb package source files

▼ Assets 6

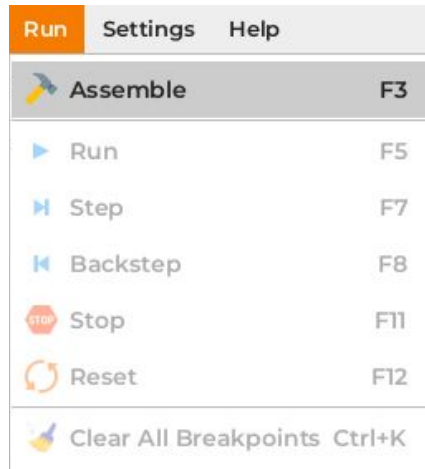
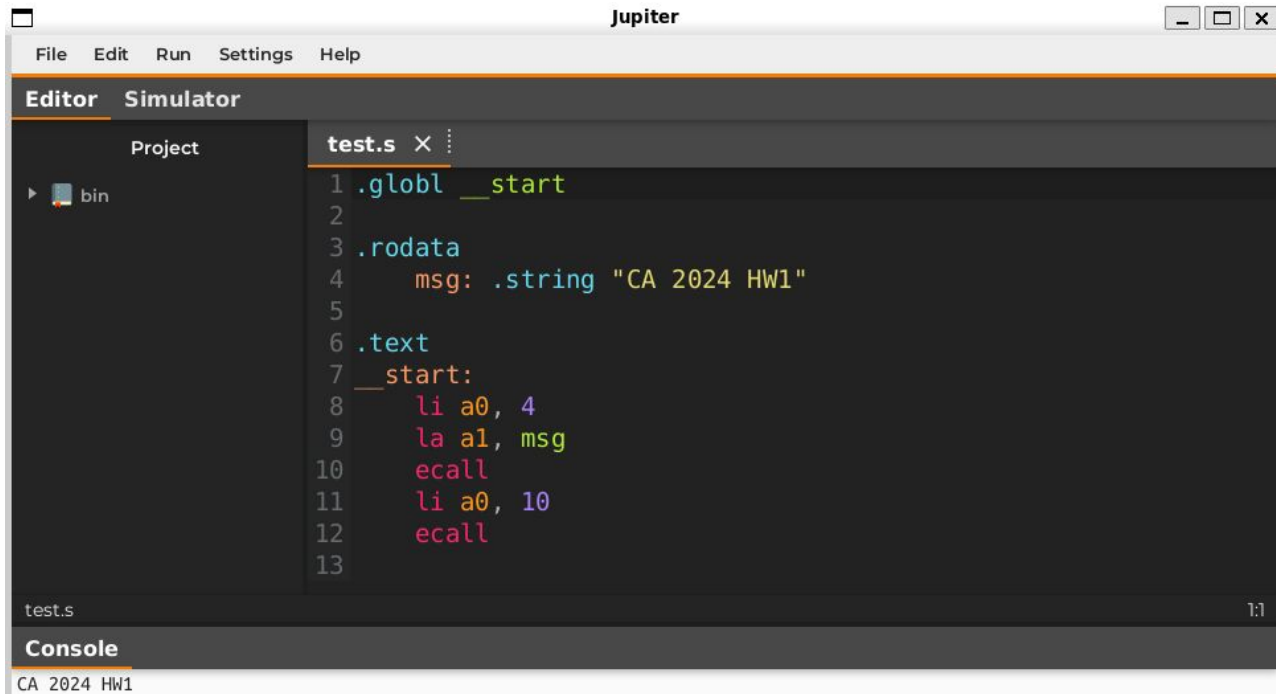
 Jupyter-3.1-linux.zip	40 MB	Sep 4, 2019
 Jupyter-3.1-mac.zip	35.7 MB	Sep 4, 2019
 Jupyter-3.1-win.zip	37.6 MB	Sep 4, 2019
 jupyter_3.1_amd64.deb	87.3 KB	Sep 4, 2019
 Source code (zip)		Sep 4, 2019
 Source code (tar.gz)		Sep 4, 2019

Jupyter GUI

- Extract the downloaded file, execute image/bin/jupyter.



Jupiter GUI



Jupiter GUI

Jupiter

File Edit Run Settings Help

Editor Simulator

Bkpt	Address	Machine Code	Basic Code	Source Code
☐	0x00010000	0x00000317	auipc x6, 0	auipc x6, 0
☐	0x00010004	0x00830067	jalr x0, x6, 8	jalr x0, x6, 8
☐	0x00010008	0x00400513	addi x10, x0, 4	li a0, 4
☐	0x0001000c	0x00000597	auipc x11, 0	la a1, msg
☐	0x00010010	0x01458593	addi x11, x11, 20	la a1, msg
☐	0x00010014	0x00000073	ecall	ecall
☐	0x00010018	0x00a00513	addi x10, x0, 10	li a0, 10
☐	0x0001001c	0x00000073	ecall	ecall

Registers			Memory	Cache
Mnemonic	Number	Value		
zero	x0	0x00000000		
ra	x1	0x00000000		
sp	x2	0xbffffff0		
gp	x3	0x10000000		
tp	x4	0x00000000		
t0	x5	0x00000000		
t1	x6	0x00010000		
t2	x7	0x00000000		
s0	x8	0x00000000		
s1	x9	0x00000000		
a0	x10	0x0000000a		
a1	x11	0x00010020		
a2	x12	0x00000000		

Console

CA 2024 HW1

Jupiter CLI

- Extract the downloaded file, execute image/bin/jupiter with your code.

```
> ./jupiter hw1.s
10
0
10
20
Jupiter: exit(0)
```

Add

```
> ./jupiter hw1.s
10
3
0
division by zero
Jupiter: exit(0)
```

Division

```
> ./jupiter hw1.s
2
5
5
32
Jupiter: exit(0)
```

Power

Assignment Definition

Requirement

The calculator should support the following operations:

$+$, $-$, $\times$, $/$, $\min$, $^$, $!$

Input format:

A

$operator$

B

Output format:

[Result]

$0 \leq A, B \leq 1024, op \in \{0, 1, 2, 3, 4, 5, 6\}$

If $op = 0$, calculate $A + B$ and output the result.

If $op = 1$, calculate $A - B$ and output the result.

If $op = 2$, calculate $A \times B$ and output the result.

If $op = 3$, calculate A / B and output the result. (Quotient)

If $op = 4$, calculate minimum (A, B) and output the result.

If $op = 5$, calculate A^B and output the result.

If $op = 6$, calculate $A!$ and output the result. (In this case, $B = 0$)

If division by zero occurs, the program should print "*division by zero*".

(Don't worry about overflow or underflow.)

Note : No need to handle 0^0 .

Input

Every input file has three lines. The first line contains a non-negative integer A , the second line contains a non-negative integer op , the third line contains a non-negative integer B , corresponding to the first operand, the operator, and the second operand.

Output

The output should contain only one integer that is the result of the input equation.